

Mathematical Foundations of Political Methodology

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November 14, 2019

Conditional Expectation as an Estimator

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So we know $\hat{Y}$ is an estimator.

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This estimator's error centers on 0.

Correlation of Estimator and Estimation Error

Suppose $\hat{Y}$ is the estimator from before, and $\tilde{Y}$ the estimation error.

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The error is uncorrelated with the estimate.

Law of Total Variance

Suppose X and Y are random variables. Then

$$\text{var}[X] = E[\text{var}[X|Y]] + \text{var}(E[X|Y])$$

The first term is the average of the variation for each value of Y .

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Law of Total Variance: application

Suppose θ is a parameter, and x is data.

$$\text{var}[\theta] = E[\text{var}[\theta|x]] + \text{var}(E[\theta|x])$$

The expected variance of θ conditional on data (the posterior) is less than the variance in θ without the data (the prior).

Change of Coordinates

Proposition

Suppose X is a random variable and $Y = g(X)$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ that is a monotonic function.

Define $g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ such that $g^{-1}(g(X)) = X$ and is differentiable. Then,

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{\partial g^{-1}(y)}{\partial y} \right| \text{ if } y = g(x) \text{ for some } x \\ &= 0 \text{ otherwise} \end{aligned}$$

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Then this is a pdf because $\frac{\partial g^{-1}(y)}{\partial y} > 0$.

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$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{\partial g^{-1}(Y)}{\partial y} \right| \\ &= f_X(y^{1/n}) \frac{y^{\frac{1}{n}-1}}{n} \end{aligned}$$

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- General Normal distribution
- Chi-Squared Distribution

Getting to Central Limit Theorem

- Use inequalities to prove Weak Law of Large Numbers (WLLN)
- Use Taylor's Theorem to show how a Moment Generating Function (MGF) works.
- Use MGF and WLLN to prove Central Limit Theorem.

Inequalities and Limit Theorems

Limit Theorems

- What happens when we consider a long sequence of random variables?
- What can we reasonably infer from data?
 - Laws of large numbers: averages of random variables converge on expected value?
 - Central Limit Theorems: sum of random variables have normal distribution?

Weak Law of Large Numbers

Proof plan:

- Markov's Inequality
- Chebyshev's Inequality
- Weak Law of Large Numbers

Markov's Inequality

Proposition

Suppose X is a random variable that takes on non-negative values. Then, for all $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

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Proof.

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For $a > 0$,

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Because $X \geq 0$,

$$E[X] \geq \int_a^{\infty} xf(x)dx \geq \int_a^{\infty} af(x)dx = aP(X \geq a)$$

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$$\begin{aligned} E[X] &\geq \int_a^{\infty} xf(x)dx \geq \int_a^{\infty} af(x)dx = aP(X \geq a) \\ \frac{E[X]}{a} &\geq P(X \geq a) \end{aligned}$$



Chebyshev's Inequality

Proposition

If X is a random variable with mean μ and variance σ^2 , then, for any value $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Chebyshev's Inequality

Proof.

Define the random variable

$$Y = (X - \mu)^2$$

Where $\mu = E[X]$.

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Applying the inequality:

$$\begin{aligned} P(Y \geq k^2) &\leq \frac{E[Y]}{k^2} \\ P((X - \mu)^2 \geq k^2) &\leq \frac{E[(X - \mu)^2]}{k^2} \end{aligned}$$

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Example of Markov and Chebyshev

Suppose that we know that the number of items produced in a factory is a random variable with mean 500.

- How likely are we to see a production of 1000?
- If the variance is 50, how likely is the production between 400 and 600?

Solution

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- Markov Solution:

$$P(X \geq 1000) \leq \frac{E(X)}{1000} = \frac{500}{1000}$$

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- If the variance is 50, how likely is the production between 400 and 600?
- Chebyshev Inequality (set k to 100, $\sigma^2 = 50$):

$$P(|X - 500| \geq 100) \leq \frac{\sigma^2}{(100)^2} = \frac{50}{10000} = \frac{1}{200}$$

$$P(|X - 500| < 100) \geq 1 - \frac{1}{200}$$

Random Variables, Data and the True parameters

- Capital letters X, Y, Z are random variables.
- Greek letters $\mu, \sigma, \alpha, \beta, \lambda, \theta, \delta$ are the True parameters.

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- Observed facts about the world are not the TRUTH - we see only shadows!
 - We infer the true value by observing data, but we can only approximate the truth.
 - e.g. Fundamental problem causal inference $\delta = Y_t(i) - Y_c(i)$
- To recognize this, we model our data as random variables, summaries of which are also random variables.

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- $\bar{X} \neq \mu$

Sequences of Random Variables

Sequence of Independent and Identically Distributed Random variables.

- Sequence: $X_1, X_2, \dots, X_n, \dots$
- Think of a sequence as sampled **data**:
 - Suppose we are drawing a sample of N observations
 - Each observation will be a random variable, say X_i
 - With realization x_i

Mean/Variance of Sample Mean

Proposition

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and variance σ^2 . Let $\bar{X}_n$ be the sample mean. Then $E[\bar{X}_n] = \mu$ and $\text{var}(\bar{X}_n) = \frac{\sigma^2}{n}$

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Proof.

$$E[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^n E[X_i]$$

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Proof.

$$\begin{aligned} E[\bar{X}_n] &= \frac{1}{n} \sum_{i=1}^n E[X_i] \\ &= \frac{1}{n} n\mu = \mu \end{aligned}$$



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Weak Law of Large Numbers

Proposition

Suppose $X_1, X_2, \dots, X_n$ is a random sample from a distribution with mean μ and $\text{Var}(X_i) = \sigma^2$. Then, for all $\epsilon > 0$,

$$P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \mu \right| \geq \epsilon \right\} \rightarrow 0 \text{ as } n \rightarrow \infty$$

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From our previous proposition

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Further,

$$E\left[\left(\frac{\sum_{i=1}^n X_i - n\mu}{n}\right)^2\right] = \frac{\text{Var}(X_1 + X_2 + \cdots + X_n)}{n^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2} = \frac{\sigma^2}{n}$$

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Apply Chebyshev's Inequality:

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Apply Chebyshev's Inequality:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| \geq \epsilon\right\} \leq \frac{\sigma^2}{n\epsilon^2}$$



Suppose $X_1, X_2, \dots$ are iid normal distributions,

$$X_i \sim \text{Normal}(0, 10)$$

$$P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \mu \right| \geq 0.1 \right\} \text{ as } n \rightarrow \infty$$

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$$0.01 = \frac{10}{n(0.1^2)}$$

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$$\begin{aligned} 0.01 &= \frac{10}{n(0.1^2)} \\ n &= \frac{1000}{0.01} \end{aligned}$$

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$$\begin{aligned} 0.01 &= \frac{10}{n(0.1^2)} \\ n &= \frac{1000}{0.01} \\ n &= 100,000 \end{aligned}$$